

The Impact of an 8-Week Resisted Sprint Training Program on Ice Skating Performance in Male Youth Ice Hockey Players

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Abstract

Dietze-Hermosa, MS, Montalvo, S, Gonzalez, MP, and Dorgo, S. The impact of an 8-week, resisted, sprint training program on ice skating performance in male youth ice hockey players. *J Strength Cond Res* 38(5): 957–965, 2024—The purposes of this randomized control study were to (a) compare the effects of an on-ice versus an overground resisted sprint training intervention and a control condition and (b) identify changes in ice skating kinematics and kinetics after training intervention participation. Twenty-four youth ice hockey players were randomly allocated into 3 groups: (a) on-ice resisted sprint training (on-ice RST); (b) overground resisted sprint training (overground RST); and (c) body weight resistance training (control). During the 8-week intervention, the 2 RST groups engaged in sled towing methods, whereas the control group engaged in a body weight resistance training program twice a week. A series of individual, repeated-measures analysis of variances with post hoc pairwise comparisons were conducted for variables of interest. An interaction effect was noted for ice skating s-cornering agility drill completion time ($p = 0.01$; $\eta_p^2 = 0.36$), ice skating 30-m top speed completion time ($p = 0.04$; $\eta_p^2 = 0.27$), step length ($p = 0.04$; $\eta_p^2 = 0.26$), and knee angle at touchdown ($p = 0.03$; $\eta_p^2 = 0.30$). The on-ice RST group displayed superior improvements across ice skating tests compared with the control group. Data show that on-ice RST has the greatest transfer effect to ice skating metrics; however, improvements in certain ice skating metrics can be observed with overground training also.

Key Words: speed, ice hockey, sled training, kinematics, force-velocity profile

Introduction

Ice hockey requires athletes to rapidly skate up and down the ice rink to contribute to an offensive or defensive play. Maximal ice skating speed during straight-line ice skating is one of the main contributors to successful ice hockey performance (33). Moreover, ice skating acceleration is needed to rapidly change directions during gameplay (33,36). Intuitively, ice skating speed and acceleration aids players during game-defining moments. Thus, the opportunities to either prevent or generate an important play are more numerous for faster ice hockey players compared with their slower counterparts. Robbins et al. (36) found that high-caliber players displayed an average speed of $6.00 \text{ m}\cdot\text{s}^{-1}$ compared with $5.53 \text{ m}\cdot\text{s}^{-1}$ for low-caliber players. Therefore, increasing maximal ice skating speed and acceleration is often a main goal of ice hockey strength and conditioning training programs.

Strength and conditioning programs for ice hockey players have primarily included overground (land-based) training. For instance, explosive exercises such as Olympic lifts (clean, snatch) are widely incorporated into training programs by ice hockey strength and conditioning coaches with a desired translation to

superior ice skating acceleration (11). Overground training studies adopting high-velocity, high-load, and strength-based exercises have resulted in improved ice skating acceleration and maximum ice skating speed (7,12,28). An 8-week combined plyometric and strength program was effective at reducing 10-m ice skating acceleration completion time by 2.8% ($p = 0.03$) in high-level ice hockey players (7). Moreover, Naimo et al. (28) reported a 5.2% decrement in maximal 33-m ice skating completion time in a high-intensity training group compared with only a 1.2% decrease in the low-intensity training group. Yet, only a few studies exist that explored the impact of any on-ice training for ice hockey players.

Few studies have incorporated an on-ice approach to training; some being acute in nature (25), of a very short (4-week) intervention duration (19) or implementing a weighted puck as the resistance implement (29). Matthews et al. (25) explored the impact of a single session of resisted ice skating on subsequent unresisted ice skating speed, concluding that a single bout of resisted ice skating decreased 25-m ice skating completion time by 2.6% ($p = 0.02$) in collegiate, male, ice hockey players (25). Janot et al. (19) reported that in youth (11–14 years old) ice hockey players, a short (4-week; twice a week), resisted, ice skating training program reduced 45-m ice skating completion time by 4.3% ($p = 0.03$) and 6-m ice skating acceleration completion time by 3.4% ($p = 0.04$). As acknowledged by both authors, the variable resistance provided by the elastic cords does not enable accurate quantification of the resistance because the resistance will depend on the effort of the subject, the elastic cord

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tension, and the effort of the partner holding the elastic cord (19,25). Consequently, the resistance experienced by the subject is unknown, and players can experience distinct resistances when elastic cords are used. Moreover, only Janot et al. (19) explored the impact of multiple sessions of resisted ice skating, given a 4-week study duration, the authors suggested that longer interventions should be explored to properly assess the effects on skating performance.

Thus, currently, there is a paucity of research exploring the impact of on-ice training with a known resistance on ice skating speed and acceleration in ice hockey players. By contrast, various studies have explored the impact of training modalities with a known resistance to enhance running speed and acceleration in overground sports (1). One such modality is sled towing, a form of resisted sprint training that may lead to improvements in sprint acceleration and maximum speed as reported across athletes of different overground sports (1). In addition, the limited research on ice hockey training have frequently omitted relevant physiological and biomechanical measures. Despite certain differences in kinematics between ice skating and overground sprinting, both ice skating and overground sprinting rely heavily on lower-limb range of motion and musculature (2,18). Moreover, both necessitate the proper application of horizontal forces to attain maximal speeds and rapid acceleration (10,31). Nevertheless, there is a gap in the literature exploring the effects of a training intervention on ice skating kinematics in ice hockey players.

Considering the sprint performance enhancing effects of resisted sprint training in overground sports, it is hypothesized that a longitudinal resisted sprint training program would also improve ice skating speed and acceleration. However, it is unknown whether an on-ice resisted sprint training program is superior to an overground resisted sprint training program to increase ice skating speed and acceleration. Thus, the purpose of this quantitative study was to compare the effects of 3 distinct training programs on ice skating measures in ice hockey players. The aims of this study were (a) to investigate the effects of an on-ice resisted sprint training intervention, an overground resisted sprint training intervention, and a no supplemental sprint training control group condition on maximal ice skating speed, acceleration, and agility and (b) to identify changes in ice skating kinematics and kinetics during maximal ice skating after participation in these different intervention conditions. It was hypothesized that both intervention groups would improve ice skating speed, acceleration, and agility beyond any improvements displayed by the control group. It was also hypothesized that ice skating kinematics and kinetics would be altered by participation in either resisted sprint training intervention.

Methods

Experimental Approach to the Problem

A randomized control trial was implemented. Athletes were recruited from a local high school ice hockey team and were randomly assigned to 8 weeks of either on-ice or overground resisted sprint training or a pseudocontrol group. All athletes underwent a familiarization session where they practiced and performed all testing and training procedures. During the subsequent session, athletes completed pretesting (baseline) measures consisting of a series of ice skating tests. During the on-ice acceleration sprints, split times, kinematics, and kinetic data were collected. Following pretesting, the randomized intervention groups participated in 2 resisted sprint training sessions per week on nonconsecutive days for 20–30 minutes following

recommendations from the previous literature (1). After the 8-week intervention, posttesting took place following the same design as outlined for pretesting.

Subjects

The ice hockey team consisted of players ages 14–18 years old. The team's roster consists of 30 players, and it was anticipated that all players would participate in the study. The study was approved by the Brigham Young University-Idaho Institutional Review Board before the commencement of data collection. Signed parental consent and subject assent forms were collected by the research team. Subjects were informed of the benefits and risks of the investigation before signing the institutionally approved document.

Using the G*Power software (version 3.1, Universität Kiel, Kiel, Germany), a repeated-measures analysis of variance (ANOVA) a priori analysis was conducted using 35-m ice skating completion time data from Farlinger et al. (12) (Cohen's $f = 0.34$), an α of 0.05, statistical power of 0.80, indicating that a sample of 24 players (8 per group) was needed. The recruitment of 30 subjects aimed to account for potential dropouts and withdrawals from the study. Over the course of the study, 6 subjects did not complete the study because of illness ($n = 3$), injury ($n = 1$), or relocation ($n = 2$). This resulted in a final sample size of 24 players.

Procedures

Anthropometrics. Baseline subject's characteristics, such as age (years), height (in meters), and body mass (in kilograms) were obtained.

On-Ice 36.58-m Acceleration Sprints. Subjects completed 3 maximal 36.58-m (40-yard) sprints separated by a 2-minute rest. Timing gates (TG; TC-timer, Brower timing system, Draper, UT) captured completion time during these skating sprints. The trial commenced with subjects adopting an athletic stance at the 0-m mark, with the remote starter placed at the rear-leg heel of the athlete. Time commenced once subjects moved their heel from the remote starter and continued until they crossed the TG placed at 36.58 m (Figure 1A). Time was recorded to the nearest 100th of a second. Similar protocols have been previously implemented in other studies assessing ice hockey players (7,34). An additional set of TG were placed at the 9.14-m (10-yard) mark to capture split times. The 36.58-m sprint distance was selected because it mimics the distance covered during a prolonged bout of offensive or defensive on-ice rush and previously has been implemented in ice hockey research (13).

On-Ice 30-m Top Speed Sprints. Subjects additionally completed 3 maximal top speed sprints separated by a 2-minute rest. Timing gates were used to capture completion time. The trial commenced with subjects adopting an athletic stance and at the opposite side of the ice rink at the red line. Athletes slowly gathered speed and continued around the net area, whereupon they progressed to top speed. Subjects were at top speed once reaching the first marker (0-m mark) and attempted to maintain top speed until the end marker set at a 30-m distance from the first marker (Figure 1B). Time commenced once subjects crossed the first marker (timing gate) and continued until they crossed the opposite TG placed 30 m away and was recorded to the nearest 100th of a second. A

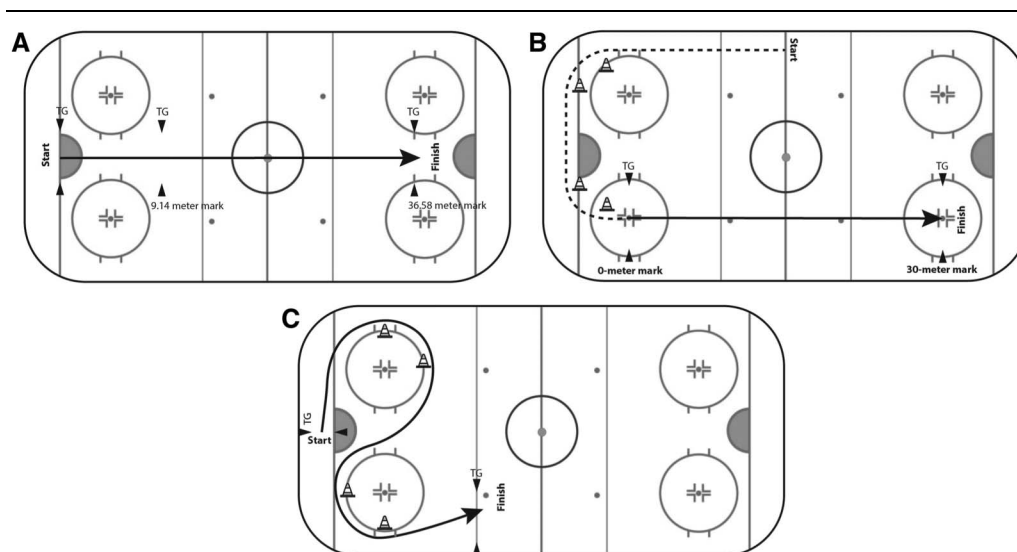


Figure 1. (A) Ice skating 36.58-m acceleration phase test. (B) Ice skating 30-m top speed phase test. (C) On-ice s-coring agility drill test. TG = timing gates.

similar protocol was implemented in previous studies aiming to assess ice skating top speed in ice hockey players (19,20).

On-Ice S-Cornering Agility Drill Test. Subjects completed 3 maximal s-coring agility tests separated by a 2-minute rest. Timing gates were used to capture completion time. The trial commenced with subjects adopting an athletic stance behind the goal line and net. Athletes maximally ice skated around the 2 near face-off circles in an S-type fashion (Figure 1C). Time commenced once subjects moved their heel from the remote starter and continued until they crossed the TG placed at the blue line. Time was recorded to the nearest 100th of a second. If a player cut inside the face off circles or fell, the trial was restarted. Cones were placed around the face-off circles to enforce proper ice skating movement during testing. Although it is acknowledged that this test aligns with change of direction over true agility, the protocol has been implemented to assess ice skating “agility” completion time in ice hockey players (20,30).

Force-Velocity Skate Profile. During the on-ice acceleration sprint, the subject’s skate (or sprint) profile was obtained. This enabled the creation of the force-velocity and power-velocity curves of athletes during sprinting (5,9,10,38). In brief, the simple method allows for the calculation of theoretical maximal force ($\text{N}\cdot\text{kg}^{-1}$), velocity ($\text{m}\cdot\text{s}^{-1}$), and peak power ($\text{W}\cdot\text{kg}^{-1}$). The details of the methodology can be found elsewhere (38). Recent research suggested a practical method to obtain the force-velocity and power-velocity curves using a single high-speed camera system (240 fps) and markers placed at specific distances (5, 10, 15, 20, 25, and 30 m) (37). Moreover, this methodology was used to reliably obtain sprinting profiles during maximal ice skating in hockey players (31,32,40). Therefore, a single high-speed camera, markers placed at specific distances, the associated mobile application (My Sprint), and corresponding Microsoft Excel spreadsheet were used to obtain the force-velocity profile of subjects during on-ice 36.58-m acceleration skating. An iPad Air (Apple Inc., Cupertino, CA, 240 fps) was used for data collection. The iPad was placed on a fixed tripod at a height of 1.5 m and a perpendicular distance of 10 m from the 15-m mark as outlined in the application and corresponding literature (37,40).

Skating Kinematics. The study adopted similar approaches as prior literature to obtain kinematics during ice skating acceleration sprints (8,43). In brief, a single high-speed camera (iPad Air, Apple Inc., Cupertino, CA) with a capturing frequency of 240 fps was placed on a stationary stand with the optical axis perpendicular to the movement plane of the lower limbs during ice skating, enabling accurate identification of joint angular kinematic parameters and spatiotemporal parameters of interest. The captured video file was imported into the Kinovea software (v.0.9.5) for kinematic analysis. Recently, sprint and resisted sprint literature have used the Kinovea software to obtain relevant kinematic parameters (23,43,44). The details of the method used to calculate kinematic parameters, such as joint angles, can be found elsewhere (8,43). Joint angles were calculated during toe-off and touchdown of the right limb within the 10- to 20-m range during the on-ice 36.58-m acceleration sprint.

To ascertain the spatiotemporal variables of interest (flight time, contact time, step length, step frequency), the touchdown (first frame when the foot is visibly in contact with the ground) and toe-off (first frame when the foot has visibly left the ground) were identified. Importantly, calculation of flight time was modified as there exists no “true” flight phase during ice skating given that after the first few steps, one skate blade always remains in contact with the ice (3). Therefore, flight time was calculated as the time from toe-off to the subsequent touchdown of the same limb, representing a stride rather than a step (36,39). The first 2 steps in the 10- to 20-m distance using the 6× zoom in Kinovea were selected (23,43). Each video was analyzed twice by the same observer, and the same leg sequence was analyzed for pretesting and posttesting. The video footage was calibrated using a pike (1.2 m) located vertically at the 15-m mark in the center of the camera field of view. The digitized coordinates were used to obtain contact time (s), flight time (s), step length (m; the horizontal displacement between subsequent touchdowns), and step frequency (Hz; the reciprocal of step duration, determined as the sum of step contact time and subsequent flight time) (23,43,44).

Training Program. Athletes were randomly assigned to 1 of 3 groups: (a) an on-ice resisted sprint training intervention, (b) an

overground resisted sprint training intervention, or (c) a pseudo control group. The pseudo control group consisted of athletes who performed their team-assigned strength and conditioning regimen. This regimen remained unaltered; hence, the designation of pseudo control. However, for the sake of brevity and simplicity, this group will be referred to simply as the “control group” throughout the article. The control group was engaged in the prescribed training program of the sport coaching staff because the omission of any training would likely induce detraining effects and result in decreased physical performance (see Table 2, Supplemental Digital Content 1, <http://links.lww.com/JSCR/A460>).

The on-ice intervention was completed in a standard dimension ice hockey rink on freshly Zambonied ice. The overground resisted sprint training and control groups completed the training interventions in a local school gym with finished hardwood flooring. The overground resisted sprint training group participated in 2 weekly sessions of 6–9 maximal sprints pulling a sled over a 20-m distance, with subjects given 3 minutes of rest between each resisted sprint. The on-ice resisted sprint training group participated in 2 weekly sessions of 6–9 maximal skating sprints over a 20-m distance (see Table 1, Supplemental Digital Content 1, <http://links.lww.com/JSCR/A460>). All training groups completed the training without wearing full ice hockey gear. For safety, the on-ice group wore helmet, gloves, and ice hockey stick in addition to their ice skates. The sled was used for the resisted sprint interventions as suggested by ice hockey strength and conditioning practitioners (35). The athletes were harnessed to the sled through a nonelastic nylon tether attached to the sled by high-tensile karabiners. The sled was loaded, accounting for the unloaded sled weight, with calibrated weight plates to achieve the corresponding load needed for the athlete (Figure 2). The sled load for the resisted sprint (both for overground and on-ice) was determined through load-velocity testing; a detailed explanation of the methodology can be found elsewhere (6,23). In brief, the force-velocity relationship during sprinting using multiple trials of resistive loads is reported to be linear (5). Thus, athletes completed 5 sprints (30-m unresisted; 30-m at 25%; 20-m at 50%; 20-m at 75%; and 15-m at 100% body mass), which allowed for an associated linear load-velocity profile to be created for each athlete. This load-velocity profile represents a span of loading under which certain mechanical determinants, including optimal power, can be targeted (6). The optimal conditions for power were calculated at the point of maximal theoretical power, corresponding

to $0.5 \times$ maximal theoretical velocity and $0.5 \times$ maximal theoretical horizontal force (6). Thus, the velocity at which maximal power was produced was then substituted into the load-velocity equation to provide the loading that corresponded to this velocity (6). This resulted in an individualized sled load for each athlete participating in either the on-ice resisted sprint or overground resisted sprint training programs. The specific training program details (distance, frequency, volume, and duration) were based on recent resisted sprinting literature with team-based athletes (1,23).

The session duration for both intervention groups lasted 45–60 minutes depending on the week. Before each training session, subjects engaged in a 10-minute standardized warm-up, including submaximal unresisted sprints and body weight movements targeting major muscle groups of the upper and lower extremities. Following the warm-up, the subjects engaged in their assigned training program. The athletes did not engage in any additional strength and conditioning program outside of their assigned study intervention (on-ice = only resisted ice skating; overground resisted sprinting = only resisted sprinting; control = only body weight exercise program) for the duration of the study intervention. All players participated in 2 to 3 weekly ice hockey practices or games and 1 to 2 60-minute weekly physical education class in their high school.

Statistical Analyses

All data collected for variables of interest were restructured in a comprehensive Microsoft Excel spreadsheet and imported into JASP software (JASP Team [2023]. JASP (Version 0.17.3) [Computer software]) based in R statistical programming language for statistical processing. Assumptions of normality were verified using the Shapiro-Wilk tests and visual assessment of distribution through histogram. The test-retest reliability of each variable of interest was determined using intraclass correlation coefficient (ICC, 2-way mixed effects, absolute agreement) with 95% confidence intervals (CIs). Intraclass correlation coefficients were interpreted as follows: <0.40 = poor; 0.40 – 0.60 = fair; 0.60 – 0.75 = good; 0.75 – 1.00 = excellent (21). The coefficient of variation (CV) was also calculated using a Microsoft Excel spreadsheet with acceptable reliability for outcome measures having an ICC of $>80\%$ and CV of $\leq 10\%$ (16).

Most variables of interest displayed acceptable test-retest reliability; thus, the average value was used for statistical analysis (see Table 3, Supplemental Digital Content 1, <http://links.lww.com/JSCR/A460>). Levene's test was used to ensure that the data met the criteria for homogeneity of variance. To establish the effects of the intervention on outcome variables, a series of repeated-measures ANOVA with 2 time points and 3 groups as fixed factors were performed. Significant interactions were decomposed using Holm-Bonferroni adjusted pairwise comparisons. When assumptions of sphericity were violated, the Huynh-Feldt correction was reported. Moreover, partial eta square (η_p^2) calculated from the repeated-measures ANOVAs were provided and interpreted as follows: small ($\eta_p^2 = 0.01$); medium ($\eta_p^2 = 0.09$); and large ($\eta_p^2 = 0.25$) effects (24). Mean differences and 95% CIs between pretesting and posttesting for outcomes measures were also calculated (Tables 1–4).

In addition, Cohen's d was calculated to represent effect sizes using pooled SD in a custom Microsoft Excel spreadsheet. The effect sizes were interpreted as follows: trivial = ≤ 0.20 ; small = 0.20 – 0.60 ; moderate = 0.60 – 1.2 ; large = 1.2 – 2.0 ; very large = 2.0 – 4.0 ; nearly perfect >4.0 (17). All data were analyzed at a significance level of 0.05.



Figure 2. Example of the on-ice resisted sprint training.

Table 1
Athlete descriptive information by group and time.*

	Pretesting						Posttesting						Pairwise comparison		
	Control (n = 8)		Overground (n = 8)		On-ice (n = 8)		Control (n = 8)		Overground (n = 8)		On-ice (n = 8)		Control	Overground RST	On-ice RST
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD			
Height (m)	1.72	0.10	1.78	0.04	1.71	0.11	1.74	0.12	1.78	0.04	1.72	0.13	−0.02 (−0.08, 0.00)	0.00 (−0.09, 0.00)	−0.01 (−0.10, 0.00)
Mass (kg)	65.66	10.89	69.48	8.01	59.67	17.90	66.19	11.21	69.98	8.35	60.43	17.39	−0.53 (−1.11, 0.00)	−0.50 (−1.26, 0.00)	−0.86 (−1.45, 0.00)
BMI (kg·m ^{−2})	22.06	2.59	21.85	2.23	20.04	3.75	22.10	2.42	22.06	2.21	20.74	3.79	−0.04 (−0.09, 0.00)	−0.21 (−0.42, 0.00)	−0.70 (−1.32, 0.00)
Age (y)	15.63	1.23	16.50	0.89	15.15	0.86	—	—	—	—	—	—	—	—	—
Sled load (kg)	—	—	34.10	4.54	43.10	6.80	—	—	—	—	—	—	—	—	—

*BMI = body mass index; CI = confidence intervals.

Results

Ice Skating Completion Time

There were significant group $\times$ time interaction effects on ice skating s-cornering agility drill completion time ($F(2, 21) = 5.634$; $p = 0.011$; $\eta_p^2 = 0.360$; large) and ice skating 30-m top speed completion time ($F(2, 21) = 3.781$; $p = 0.040$; $\eta_p^2 = 0.274$; large; Figure 3A,B). Follow-up analyses indicated significant pretesting to posttesting changes for the on-ice RST group on ice skating s-cornering completion time (mean diff = -0.412 seconds; $p < 0.001$; Cohen's $d = 1.75$; large). The pre-to-post changes for both the overground RST and control group did not reach statistical significance for the s-cornering agility drill or ice skating 30-m top speed completion times ($p > 0.05$). Although not significant at the 0.05 level, there was a moderate group $\times$ time interaction for 36.58-m completion time ($F(2, 21) = 2.854$; $p = 0.081$; $\eta_p^2 = 0.174$; moderate). Follow-up analyses revealed pretesting to posttesting changes for the on-ice RST group (mean diff = -0.847 seconds; $p < 0.001$; Cohen's $d = 2.43$; very large) and the overground RST group (mean diff = -0.699 seconds; $p < 0.001$; Cohen's $d = 1.69$; large) for 36.58-m completion time (Figure 3C). Finally, there was a main effect for time on 9.14-m completion time ($F(1, 21) = 9.601$; $p = 0.006$; $\eta_p^2 = 0.325$; large). This corresponds

to a 0.074-second difference across all groups (Cohen's $d = 0.65$; moderate; $p = 0.006$).

Kinematics

There was a significant group $\times$ time interaction effect on step length ($F(2, 21) = 3.530$; $p = 0.049$; $\eta_p^2 = 0.261$; large). Yet, follow-up analysis revealed only small effects both the overground RST (Cohen's $d = 0.33$; small) and on-ice RST (Cohen's $d = 0.31$; small) groups (Figure 3D). There was significant group $\times$ time interaction effect on knee swing flexion angle at touchdown ($F(2, 21) = 4.320$; $p = 0.028$; $\eta_p^2 = 0.302$; large) with the on-ice RST group displaying a decrease in the angle (Cohen's $d = 0.55$; small; Figure 3E). There was no significant group $\times$ time interaction effect, or main effect for group or time on any other kinematic variable of interest ($p > 0.05$).

Force-Velocity Skate Profile

There was a main effect for time on maximal ratio of force ($F(1, 21) = 36.87$; $p < 0.001$; $\eta_p^2 = 0.648$; large); theoretical maximal horizontal force ($F(1, 21) = 28.911$; $p < 0.001$; $\eta_p^2 = 0.599$; large); theoretical maximal horizontal power ($F(1, 21) = 35.765$;

Table 2
Ice skating completion times by group and time.*

	Pretesting						Posttesting						Pairwise comparison		
	Control (n = 8)		Overground (n = 8)		On-ice (n = 8)		Control (n = 8)		Overground (n = 8)		On-ice (n = 8)		Control	Overground RST	On-ice RST
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD			
Ice skating 9.14 m acceleration (s)	1.96	0.12	1.87	0.14	1.92	0.16	1.92	0.16	1.81	0.08	1.80	0.08	0.04 (−0.09, 0.17)	0.06 (−0.07, 0.19)	0.12 (−0.02, 0.26)
Ice skating 36.58 m acceleration (s)	6.10	0.39	6.08	0.48	6.33	0.43	5.66	0.47	5.40	0.29	5.52	0.19	0.44 (0.03, 0.86)	0.68 (0.26, 1.10)	0.81 (0.37, 1.25)
Ice skating 30 m fly (s)	3.83	0.29	3.86	0.33	4.13	0.38	3.91	0.34	3.81	0.40	3.91	0.26	−0.08 (−0.33, 0.18)	0.05 (−0.20, 0.30)	0.23 (−0.04, 0.50)†
Ice skating S-cornering agility drill (s)	9.81	0.55	9.79	0.44	10.24	0.74	9.56	0.35	9.44	0.49	9.23	0.34	0.24 (−0.32, 0.80)	0.35 (−0.21, 0.90)	1.01 (0.41, 1.60)†

*Mean differences = pre-post values.

†Pairwise comparison significant at $p = 0.05$ conducted after significant group $\times$ time interaction effect.

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Table 3
Ice skating kinematics by group and time.*

	Pretesting						Posttesting						Pairwise comparison		
	Control (n = 8)		Overground (n = 8)		On-ice (n = 8)		Control (n = 8)		Overground (n = 8)		On-ice (n = 8)		Control	Overground RST	On-ice RST
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean difference (95% CI)		
Ice skating CT (s)	0.26	0.04	0.28	0.03	0.27	0.02	0.25	0.04	0.27	0.02	0.26	0.03	0.02 (−0.02, 0.05)	0.01 (−0.03, 0.04)	0.01 (−0.03, 0.05)
Ice skating FT (s)	0.38	0.05	0.37	0.06	0.36	0.06	0.38	0.06	0.35	0.05	0.36	0.06	−0.01 (−0.07, 0.06)	0.02 (−0.04, 0.08)	−0.01 (−0.07, 0.06)
Ice skating step length (cm)	48.54	8.40	50.59	9.36	54.63	10.84	47.39	8.12	53.86	10.63	57.79	10.39	1.15 (−3.27, 5.58)	−3.27 (−7.69, 1.16)	−3.16 (−7.99, 1.57)†
Ice skating step rate (steps·min ^{−1})	1.58	0.17	1.55	0.15	1.61	0.16	1.62	0.18	1.61	0.14	1.63	0.17	−0.04 (−0.19, 0.11)	−0.05 (−0.20, 0.10)	−0.10 (−0.18, 0.14)
Trunk angle at TO (°)	47.95	6.97	45.28	5.73	43.91	5.66	47.11	6.84	49.44	6.15	44.61	3.89	0.84 (−4.73, 6.41)	−4.16 (−9.73, 1.40)	−0.71 (−6.66, 5.24)
Hip flexion (swing leg) at TO (°)	103.01	6.47	99.79	7.55	96.54	8.25	102.67	6.05	103.21	6.11	97.09	5.85	3.03 (−3.61, 9.67)	−2.62 (−9.26, 4.02)	−0.11 (−7.21, 6.98)
Knee flexion (swing leg) at TO (°)	110.61	5.70	110.16	9.88	108.11	7.05	108.53	7.10	107.96	7.47	109.47	5.87	−7.12 (−16.97, 2.73)	−8.44 (−18.29, 1.41)	−3.11 (−13.64, 7.42)
Hip extension (stance leg) at TO (°)	−18.21	7.57	−15.80	8.22	−9.51	12.68	−19.80	11.81	−17.32	7.59	−14.06	8.28	1.60 (−5.77, 8.96)	1.52 (−5.84, 8.88)	4.54 (−3.33, 12.41)
Knee flexion (stance leg) at TO (°)	162.92	5.95	167.49	4.18	164.98	7.95	162.60	5.05	168.59	4.65	162.10	9.77	0.32 (−4.64, 5.28)	−1.09 (−6.05, 3.86)	2.88 (−2.42, 8.18)
Trunk angle at TD (°)	53.13	6.94	48.78	6.83	47.68	5.36	52.41	6.67	51.89	7.49	47.68	2.62	0.71 (−3.43, 4.85)	−3.11 (−7.25, 1.03)	−0.01 (−4.42, 4.42)
Hip flexion (swing leg) at TD (°)	105.70	6.38	100.59	6.11	96.97	7.31	105.76	6.42	104.39	7.59	96.91	2.63	−0.06 (−4.40, 4.28)	−3.80 (−8.14, 0.54)	0.06 (−4.59, 4.70)
Knee flexion (swing leg) at TD (°)	101.41	5.10	99.52	10.93	106.36	10.93	99.26	5.92	102.65	11.59	101.13	7.87	2.16 (−4.44, 8.76)	−3.13 (−9.73, 3.47)	5.24 (−1.82, 12.29)†
Hip extension (stance leg) at TD (°)	−48.17	10.88	−32.31	17.36	−30.90	16.96	−49.51	11.12	−33.68	13.79	−34.38	14.08	1.34 (−9.32, 12.00)	1.37 (−9.29, 12.03)	3.48 (−7.92, 14.88)
Knee flexion (stance leg) at TD (°)	147.13	4.80	161.58	9.06	154.69	13.35	146.24	5.22	160.18	3.48	154.19	10.68	0.89 (−8.04, 9.82)	1.39 (−7.54, 10.32)	0.49 (−9.05, 10.04)

*CT = contact time; FT = flight time; TO = toe-off; TD = touchdown; mean differences = pre-post values.
†Pairwise comparison significant at $p = 0.05$ conducted after significant group $\times$ time interaction effect.

Table 4
Force-velocity skate profile by group and time.*

	Pretesting						Posttesting						Pairwise comparison		
	Control (n = 8)		Overground (n = 8)		On-ice (n = 8)		Control (n = 8)		Overground (n = 8)		On-ice (n = 8)		Control	Overground RST	On-ice RST
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean difference (95% CI)		
Ice skating F_0 (N·kg ^{−1})	7.45	1.43	7.96	1.72	8.02	1.68	9.42	2.04	10.27	2.45	9.90	1.75	−1.98 (−4.10, 0.15)	−2.31 (−4.44, −0.19)	−1.88 (−4.15, 0.39)
Ice skating V_0 (m·s ^{−1})	7.36	0.46	7.45	0.76	6.88	0.46	7.04	0.54	7.22	0.75	6.89	0.44	0.32 (−0.14, 0.77)	0.23 (−0.23, 0.69)	−0.01 (−0.50, 0.48)
Ice skating P_{max} (W·kg ^{−1})	13.80	3.15	14.70	3.20	13.73	3.07	16.47	3.68	18.20	3.81	16.80	2.81	−2.67 (−5.71, 0.36)	−3.50 (−6.53, −0.47)	−3.07 (−6.31, 0.17)
Ice skating S_{FV}	−1.00	0.14	−1.07	0.25	−1.15	0.23	−1.34	0.32	−1.45	0.47	−1.41	0.28	0.34 (−0.06, 0.75)	0.38 (−0.03, 0.79)	0.26 (−0.17, 0.70)
Ice skating RF_{max} (%)	40.40%	3.90%	41.40%	3.30%	39.90%	2.50%	42.90%	3.40%	43.60%	2.60%	42.60%	2.00%	−0.03 (−0.05, −0.01)	−0.02 (−0.05, 0.01)	−0.03 (−0.05, −0.01)
Ice skating D_{RF} (%)	−9.60	1.80	−10.20	2.90	−10.90	3.10	−12.50	2.60	−13.80	4.60	−13.70	2.80	0.03 (−0.01, 0.08)	0.035 (−0.01, 0.08)	0.03 (−0.02, 0.07)
Ice skating V_{opt} (m·s ^{−1})	3.68	0.23	3.74	0.38	3.44	0.23	3.55	0.28	3.60	0.38	3.45	0.22	0.136 (−0.10, 0.37)	0.13 (−0.10, 0.36)	−0.01 (−0.25, 0.24)
Ice skating speed max (m·s ^{−1})	7.04	0.45	7.13	0.65	6.63	0.42	6.88	0.49	7.00	0.66	6.73	0.42	0.16 (−0.18, 0.51)	0.12 (−0.22, 0.47)	−0.10 (−0.47, 0.27)

* F_0 = maximal theoretical force; V_0 = maximal theoretical velocity; P_{max} = maximal theoretical power; RF_{max} = maximal ratio of force; D_{RF} = decrease in ratio of force; S_{FV} = force-velocity slope; mean differences = pre-post values.

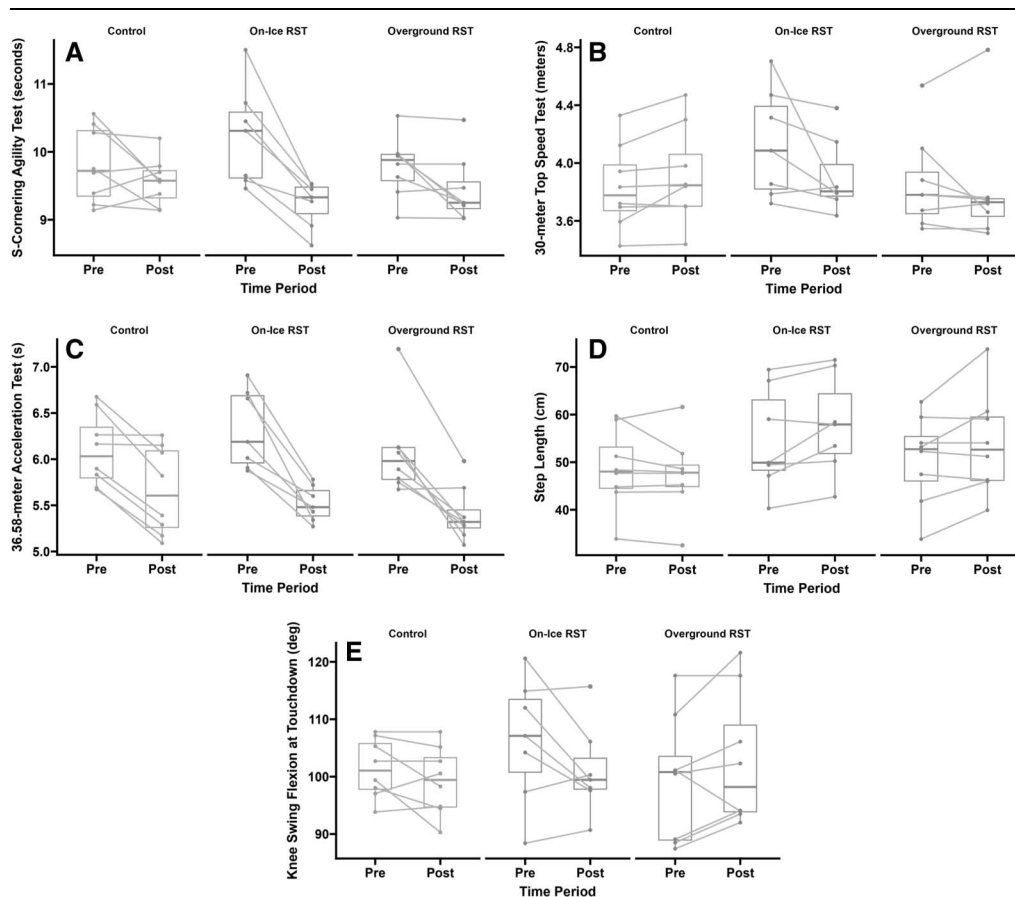


Figure 3. (A) Group by time differences in s-cornering agility test completion time. (B) Group by time differences in 30-m top speed test completion time. (C) Group by time differences in 36.58-m acceleration test completion time. (D) Group by time differences in step length. (E) Group by time differences in knee swing flexion angle at touchdown.

$p < 0.001$; $\eta_p^2 = 0.622$; large); the force-velocity slope ($F(1, 21) = 20.690$; $p < 0.001$; $\eta_p^2 = 0.508$; large); decrease in maximal ratio of force ($F(1, 21) = 17.237$; $p < 0.001$; $\eta_p^2 = 0.463$; large); and maximal speed ($F(1, 21) = 4.620$; $p = 0.044$; $\eta_p^2 = 0.188$; medium). Further analyses revealed a large increase in maximal ratio of force for the on-ice RST group (mean diff = 0.05; $p < 0.001$; Cohen's $d = 1.20$; large) and a moderate increase for the overground RST group (Cohen's $d = 0.76$; moderate). Time main effect changes correspond to a 1.972 N difference in force (Cohen's $d = 1.12$; moderate; $p < 0.001$), a 3.038 W difference in power (Cohen's $d = 1.25$; large; $p < 0.001$), a 0.328 difference in the force-velocity slope (Cohen's $d = 0.95$; moderate; $p < 0.001$), a 0.031 difference in the decrease in maximal ratio of force (Cohen's $d = 0.87$; moderate; $p < 0.001$), and a $0.06 \text{ m} \cdot \text{s}^{-1}$ difference in maximal speed (Cohen's $d = 0.13$; trivial; $p > 0.05$) across all groups. There was no interaction effect or main effect of group or time on theoretical maximal velocity or optimal velocity ($p > 0.05$).

Discussion

This study aimed to compare the effects of an 8-week on-ice resisted sprint training program, an overground resisted sprint training program, and a control body weight resistance training on ice skating completion times, ice skating kinematics, and ice skating kinetics in youth ice hockey players. The first hypothesis is supported

with decreases in ice skating test completion times noted for the on-ice RST group across all ice skating tests (9.14-m acceleration sprint, 6% decrease; 36.58-m acceleration sprint, 13% decrease; 30-m top speed, 6% decrease; and s-cornering agility drill, 10% decrease), which are superior to the control group. Furthermore, step length increased for both on-ice (5%) and overground (6%) RST during on-ice sprints. Thus, the final hypothesis is partially supported.

The current results add to the sparse literature examining the impact of on-ice training programs. Acutely, decreases in 25-m ice skating completion time (-3% ; $p = 0.02$) were reported after a single bout of on-ice elastic band RST in male ice hockey players (25). Janot et al. (19) reported an 4% increase in ice skating speed and a 4% increase in top speed among youth ice hockey players following a 4-week on-ice bungee-cord RST program. Literature reports improvements in on-ice agility after both a 4-week overground agility (2%) and a 4-week on-ice agility (3%) training programs, with the on-ice group exhibiting greater improvements (30). Therefore, the on-ice RST group study findings support those of previous studies (10% improvement in on-ice agility and 13% improvement in 36.58-m ice skating completion time). This study was 8 weeks in duration, one of the longest duration on-ice training studies, with the on-ice RST group improving on ice skating s-cornering agility drill completion time (mean diff = 1.01 seconds; $p < 0.001$; Cohen's $d = 1.75$; large; 10%; Figure 3A) and 36.58-m completion time (mean diff = 0.81 seconds; $p < 0.001$; Cohen's $d = 2.43$; very large; 13%). The overground RST group also displayed improvement in 36.58-m completion time (mean diff = 0.68

seconds; $p < 0.001$; Cohen's $d = 1.69$; large; 11%). This study presents the first insights into positive alterations in s-cornering agility drill completion time following an on-ice training program. The improvements in linear ice skating completion time of this study (13%) represent a substantial increase when compared with those reported by Janot et al. (19) (4%). The intervention duration (4 vs. 8 weeks) and the individualized resisted sled load (compared with elastic cords) may have yielded the differences observed in this study compared with those of Janot et al. During overground resisted sprint training, interventions lasting longer than 6 weeks displayed the most reduction in sprint completion times ($ES = 0.39$; $p = 0.01$) (1). Reasonably, a similar outcome may be expected during resisted ice skating. Moreover, the improvements in 36.58-m ice skating completion time exhibited by the overground RST group (11%) exceed those reported in the prior literature following an overground training intervention (2–7%) (7,12,28,30). This may be because of the increased specificity of overground resisted sprinting to maximal ice skating compared with other forms of overground training. Interestingly, it was reported that a 15-kg overground resisted sprint best predicts all-out ice skating performance in varsity ice hockey players ($r = 0.79$, $r^2 = 0.62$; $p \leq 0.001$) (41).

Improvements in completion time may be driven by alterations in the kinetic (sprint) profile components and kinematics. For instance, Perez et al. (31,32) indicated the importance of maximal horizontal power and force to attain high ice skating speeds. We found that maximal horizontal power during ice skating exhibited changes because of participating in RST. Moreover, increased trunk angle is reported in the prior literature because of RST, which may lead to increased maximal horizontal force, maximal horizontal power, and maximal ratio of force (10,23). These collective changes may explain the alterations observed in ice skating completion times.

The impact of on-ice or overground RST on ice skating kinetics is novel. Thus, findings of this study constitute the first glimpse into this new area of research. We found significant pre-to-post changes for on-ice horizontal force across all groups ([control: Cohen's $d = 1.07$; moderate; 26%], [overground RST: Cohen's $d = 1.00$; moderate; 29%], [on-ice RST: Cohen's $d = 1.04$; moderate; 23%]). Maximal ratio of force during ice skating also increased for the on-ice RST group (Cohen's $d = 1.20$; large; 7%), the overground RST group (Cohen's $d = 0.75$; moderate; 5%), and the control group (Cohen's $d = 0.68$; moderate; 6%). Therefore, participating in either an overground training program, including RST, or on-ice RST seem to alter important components of ice skating kinetics. The improvement of the on-ice RST group agrees with the principle of training specificity. As stated, the literature indicates the importance of high horizontal force, horizontal power, and maximal ratio of force to attain high ice skating speeds with research indicating RST as an effective method to alter these components (10,31). Unfortunately, given the novelty of the present work, there does not exist direct literature for comparison. In a somewhat relevant study, Haug et al. (15) explored the impact of overground sprint start training on ice skating completion time in short track speed skaters. Authors reported that a 4-week intervention performed twice a week resulted in small improvement in ice skating start time (0.07 seconds; Cohen's $d = 0.33$; small). Although kinetics were not measured, it may be that overground sprint training positively impacted ice skating performance. Both locomotion types (sprinting and maximal ice skating) require a horizontal application of force, particularly during the first few steps of the acceleration phase (22). Indeed, a positive transfer of overground body weight training and overground RST kinetics to ice skating is certainly plausible.

During ice skating sprints, there was an increase in step length for both the overground RST (Cohen's $d = 0.33$; small; 3 cm; 6%) and on-ice RST (Cohen's $d = 0.31$; small; 3 cm; 6%) groups (Figure 3D). Prior literature reports increased step or stride length because of RST, albeit during overground sprinting (1). The increase in step length may relate to the sprint (kinetic) profile component adaptations observed with the overground and on-ice RST groups during ice skating (i.e., increased maximal ratio of force and maximal horizontal power). For instance, authors propose that an increased step length may be attributed to higher impulse despite equivalent step contact time (26,27). Thus, the athlete applies more forces, particularly horizontal oriented forces, within the same time window. The on-ice RST group displayed a decrease in the knee swing flexion angle at touchdown (Cohen's $d = 0.55$; small; 5°; 5%; Figure 3E). This knee angle alteration may aid in preparing the limb for optimal contact between the ice skate blade and the ice surface, allowing the athlete to apply forces more efficiently (supporting the increase in maximal ratio of force observed in the on-ice RST group). Prior literature suggests that faster ice skaters adopt a more knee flexed position at touchdown and throughout the stance phase, which may carry over to the other ice skating improvements observed in this study (i.e., 30-m top speed and s-cornering agility drill completion times for the on-ice RST group but not for other groups) (3,4,42).

The limited strength and conditioning experience of subjects presents a limitation of the study. Thus, all athletes responded to a structured strength and conditioning program. This limitation may have impeded the study's capacity to fully elucidate the impact of different training interventions on outcome measures. In addition, it is acknowledged that the study does not include a true control group, with all groups engaging in some form of training. In this study, it was not feasible to include an inactive control group. The load used during the RST program for both groups is based on methodology used during overground RST (6). Although the load is suggested for maximal horizontal power improvements, the "best" load during on-ice RST remains to be elucidated. Therefore, researchers and practitioners seeking to implement RST with their athletes may seek to replicate this study using a variety of different loading schemes. Study findings are specific to youth male ice hockey players, and thus, extrapolating findings to other populations should be done with caution. For instance, prior research indicates ice skating kinematic differences between male and female athletes (3,39) and differences in sprint profile components between male and female athletes (14). Therefore, future randomized control studies should examine ice hockey performance metrics in female athletes.

The study only observed ice skating kinematics from a 2-dimensional perspective. This limits the overall kinematic information obtained. Ice skating research among ice hockey players is still in its infancy relative to other team-based sports. Consequently, there are many unanswered and exciting questions that researchers can undertake. Future studies should explore kinetics, kinematics, and muscle activity across a variety of ice skating movements, such as backwards and cornering ice skating. Finally, the sample size of 24 may have hindered the study's capacity to detect changes across all measured variables.

In conclusion, the training groups displayed improvements across all ice skating completion times, with the on-ice RST group having superior outcomes (30-m top speed, s-cornering agility drill) compared with the control group.

Practical Applications

Ice hockey coaches can incorporate on-ice RST to improve ice skating completion time across various ice skating scenarios. When on-ice RST is not feasible, overground training can induce positive changes across certain ice skating performance measures. Coaches may consider implementing RST as a component of a well-rounded ice hockey strength and conditioning program.

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